PERFORMANCE TEST ON HYDRAULIC RAM

AIM: To conduct performance characteristics test on the Hydraulic Ram and to find the best operating point of the Hydraulic Ram by plotting the following graphs:

1. Head Vs waste water
2. Head Vs useful water
3. Head Vs number of beats/ second
4. Head Vs Rankine's efficiency
5. Head Vs D'Aubuisson's ratio

PROCEDURE: The experiment is done at constant supply head varying the discharge pressure and useful water quantity. Open the inlet gate valve to the supply tank and maintain constant water level (2.9 m of water). The overflow pipe corresponding to the required supply head may be kept fully opened. Close the delivery valve of the Ram and start it by pressing the wastewater valve and releasing it. The delivery valve is then opened. Adjust the delivery valve of the Ram for required delivery pressure. Take the following readings; delivery pressure, weight of useful water, head causing flow in the waste water channel and time taken for 10 beats. Take six sets of readings with different delivery pressure from maximum to minimum.

SAMPLE CALCULATIONS: (READING NO.)

Delivery Head

Delivery head is measured with the help of a Pressure gauge.

$$H_d = P_1 \times 10 + z \quad \text{m of water}$$

Pressure gauge reading, P_1 = kg/cm^2

Observations and Tabular Column

Sl. No.	Pressure gauge reading P_1 kg/cm^2	Pointer gauge reading h_2 cm	Spring balance reading W_2 kg	Time for n beats t_1 s	Delivery head H_d m of water	Head causing flow h cm of water	Waste water Q m^3/s	Useful water q m^3/s	Number of beats /min	Rankine efficiency η_R %	D'Aubussion's Ratio η_D
1											
2											
3											
4											
5											
6											
7											
8											

Weight of bucket, W_1 = kg Supply Head, Hs = 2.95 m Sill level reading, h_1 = cm

Center to center vertical distance between the ram chamber and pressure gauge, z = m

Therefore, Delivery Head, H_d = x 10 +

Delivery Head, H_d =_______________m of water

Waste Water Discharge

The waste water discharge is measured using a triangular notch and its calibration equation is $Q = 0.00685 \times 10^{-3} h^{5/2}\ m^3/s$

The head causing flow is measured using a Pointer gauge or Hook gauge.

Head causing flow, $h = h_2 - h_1$ cm of water

Sill level reading, h_1 = cm

Pointer gauge (final) reading, h_2 = cm

Therefore, Head causing flow, h = - cm of water

Head causing flow, h =___________cm of water

Therefore, Waste Water Discharge, $Q = 0.00685 \times 10^{-3}$ $^{5/2}$

Waste Water Discharge, Q=___________m^3/s

Useful Water, q

The useful water is calculated by using a bucket

$$\text{Useful Water, } q = \frac{W_2 - W_1}{t \times 1000}\ m^3/s$$

Weight of bucket, W_1 = kg

Spring balance reading, W_2 = kg

(Weight of the bucket +weight of useful water)

Time, $t = 20$ s (water collected in 20 s)

Therefore, Useful Water, $q = \dfrac{-}{\times 1000}$ m^3/s

Useful Water, $q =$ ________ m^3/s

Number of Beats per minute

$$\text{Number of Beats per minute} = \frac{60 \times n}{t_1}$$

Number of beats, $n = 20$

Time for 20 beats, $t_1 =$ s

Therefore, Number of Beats per minute $= \dfrac{60 \times}{\quad}$

Number of Beats per minute= ______

Rankine efficiency

$$\text{Rankine efficiency}, \eta_R = \frac{q}{Q} \times \frac{(H_d - H_S)}{H_S}$$

Delivery Head, H_d	=	m of water
Supply Head H_S	=	m of water
Useful water, q	=	m^3/s
Waste water, Q	=	m^3/s

Therefore, Rankine efficiency, $\eta_R = \dfrac{\quad}{\quad} \times \dfrac{(\quad - 2.95)}{2.95}$

Rankine efficiency, η_R = ________ %

D'Aubussion's ratio

$$\text{D'Aubussion's ratio} = \frac{q}{(Q+q)} \times \frac{H_d}{H_S}$$

Delivery head, Hd = m

Supply head, Hs = 2.95 m

Useful water, q = m^3/s

Waste water, Q = m^3/s

Therefore, D'Aubussion's ratio, $\eta_D = \frac{\qquad}{(\quad + \quad)} \times \frac{\qquad}{2.95}$

D'Aubussion's ratio, η_D = ____________

RESULT

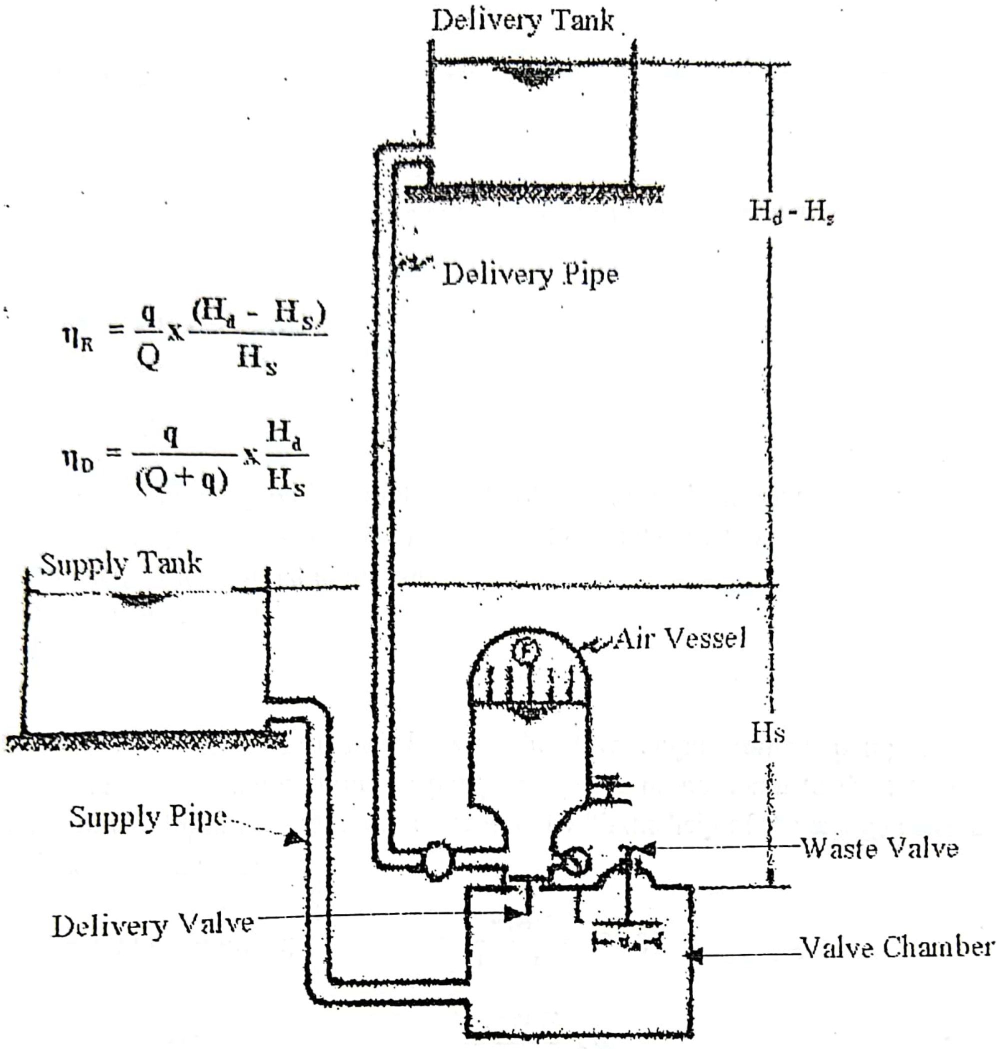
Rankine efficiency, η_R = %

D'Aubussion's Ratio, η_D =

INFERENCE

$$\eta_R = \frac{q}{Q} \times \frac{(H_d - H_s)}{H_s}$$

$$\eta_D = \frac{q}{(Q+q)} \times \frac{H_d}{H_s}$$



Schematic view of Hydraulic Ram